Features

- Integrated charge pump, support high-side charge and discharge NFET drive
- Integrated high-side pre-charge and pre-discharge PFET drivers
- Two independent ADCs
  - Support for the simultaneous voltage and current sampling
  - High-precision cell voltage measurement
  - High-precision coulomb counter
  - Support for temperature sensing using internal sensors and up to 6 external thermistors
- High-voltage tolerance of 132V on cell connect pins, and support for random cell attach sequence and hot swapping on production line
- Integrated drive for passive cell balancing
- Integrate multiple hardware protection
  - Cell over-voltage protection (COV)
  - Cell under-voltage protection (CUV)
  - Two levels of charge over-current protection (OCC1/OCC2)
  - Two levels of discharge over-current protection (OCD1/OCD2)
  - Discharge short-circuit protection (SCD)
- 3 power supply modes (typical value)
  - Normal mode: ~270μA
  - Sleep mode: ~60μA
  - Shutdown mode: ~1μA
- Supports multiple wake-up functions for sleep mode
  - I^2C communication wake-up
  - Timed scheduled wake-up
  - Charge/discharge current detection wake-up
  - 2nd level charge/discharge over-current wake-up
  - Discharge short circuit wake-up
  - The charger detects the wake-up
- 3.3V/50mA LDO is available for external systems
- 100kHz I^2C communication interface, support CRC checksum and slave address hard wire
- LQFP48(7mm×7mm) package

Applications

- Handled electric tools
- Portable energy storage system
- Uninterruptible power supply system (UPS)
- 18V，24V，36V，48V，69V，72V battery pack of Lithium iron phosphate, ternary lithium, lithium titanate, sodium ion, etc

Description

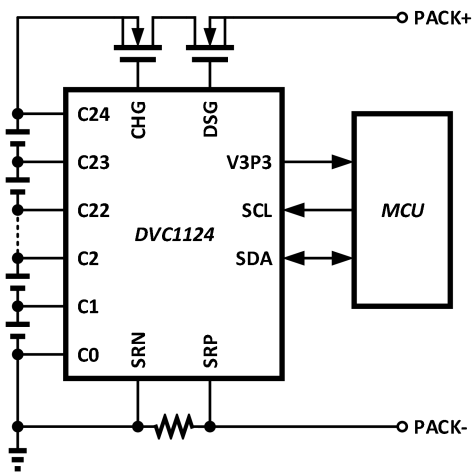
The DVC1124 is a high accuracy cell monitoring and protection IC, fabricated with vehicle specification level high-voltage BCD process, for multi-series lithium battery packs, which is suitable for the battery pack with a total voltage not exceeding 100V.

In normal mode, the VADC can complete measurements within 35ms for 24 cell voltage and 6 thermistors. While the CADC integrates 2 filters named CC1 and CC2, which output current measurement values at 4ms and 256ms respectively. In sleep mode, the external MCU system can continuously work with 3.3V LDO, and the battery pack can work in standby state with high-side charge and discharge NFET drives. Meanwhile, the 2nd level charge/discharge over-current protection and discharge short circuit protection can immediately shut down the NFET drivers in case of abnormal charging/discharging current to ensure the safety of the battery pack functions.

In sleep mode, DVC1124 supports both passive wake-up function of I^2C communication and active wake-up functions such as timer wake-up, charge/discharge current detection wake-up, 2nd level charge/discharge over-current wake-up, discharge short circuit wake-up charger detection, etc.

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Typical Application



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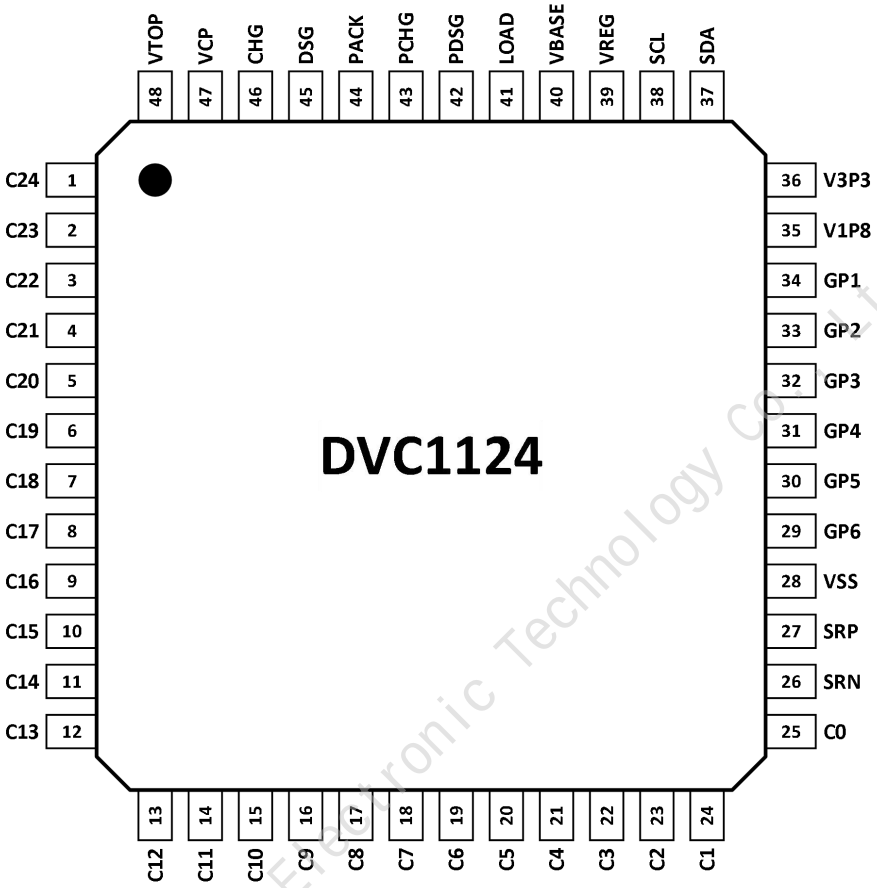
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Pin Configuration



LQFP48 top view

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Pin Function

Pin	Name	Type	Description
1	C24	I	Sensing voltage input, connected to positive terminal of the 24th cell
2	C23	I	Sensing voltage input, connected to positive terminal of the 23th cell
3	C22	I	Sensing voltage input, connected to positive terminal of the 22th cell
4	C21	I	Sensing voltage input, connected to positive terminal of the 21th cell
5	C20	I	Sensing voltage input, connected to positive terminal of the 20th cell
6	C19	I	Sensing voltage input, connected to positive terminal of the 19th cell
7	C18	I	Sensing voltage input, connected to positive terminal of the 18th cell
8	C17	I	Sensing voltage input, connected to positive terminal of the 17th cell
9	C16	I	Sensing voltage input, connected to positive terminal of the 16th cell
10	C15	I	Sensing voltage input, connected to positive terminal of the 15th cell
11	C14	I	Sensing voltage input, connected to positive terminal of the 14th cell
12	C13	I	Sensing voltage input, connected to positive terminal of the 13th cell

Pin	Name	Type	Description
13	C12	I	Sensing voltage input, connected to positive terminal of the 12th cell
14	C11	I	Sensing voltage input, connected to positive terminal of the 11th cell
15	C10	I	Sensing voltage input, connected to positive terminal of the 10th cell
16	C9	I	Sensing voltage input, connected to positive terminal of the 9th cell
17	C8	I	Sensing voltage input, connected to positive terminal of the 8th cell
18	C7	I	Sensing voltage input, connected to positive terminal of the 7th cell
19	C6	I	Sensing voltage input, connected to positive terminal of the 6th cell
20	C5	I	Sensing voltage input, connected to positive terminal of the 5th cell
21	C4	I	Sensing voltage input, connected to positive terminal of the 4th cell
22	C3	I	Sensing voltage input, connected to positive terminal of the 3th cell
23	C2	I	Sensing voltage input, connected to positive terminal of the 2th cell
24	C1	I	Sensing voltage input, connected to positive terminal of the 1th cell
25	C0	I	Sensing voltage input, connected to negative terminal of the 1th cell
26	SRN	I	Current sampling negative electrode input, connected to the sampling resistor negative terminal (near VSS)
27	SRP	I	Current sampling positive electrode input, connected to the positive terminal of the sampling resistor
28	VSS	S	GND, connected to negative terminal of the 1th cell
29	GP6	I/O	Thermistor input, analog voltage input, low-side discharge drive output, interrupt output
30	GP5	I/O	Thermistor input, analog voltage input, low-side charge drive output, interrupt output
31	GP4	I/O	Thermistor input, analog voltage input, discharge hardwired control input
32	GP3	I/O	Thermistor input, analog voltage input, low-side pre-discharge drive output, interrupt output
33	GP2	I/O	Thermistor input, analog voltage input, low-side pre-charge drive output, interrupt output
34	GP1	I/O	Thermistor input, analog voltage input, charge hardwired control input
35	V1P8	O	1.8V power output, connected to VSS through a 1μF capacitor near the pin
36	V3P3	O	3.3V power output, connected to VSS through a 1μF capacitor near the pin
37	SDA	I/O	I^2C communication bus, data input/output
38	SCL	I	I^2C communication bus, clock input
39	VREG	S	5V power input, connected to VSS through a 1μF capacitor near the pin
40	VBASE	O	NPN base drive voltage output of pre regulator, connected to VSS through a 1μF capacitor near the pin
41	LOAD	I/O	LOAD detection pin
42	PDSG	O	High-side pre-discharge drive output
43	PCHG	O	High-side pre-charge drive output
44	PACK	I	Charger detection pin
45	DSG	O	High-side discharge drive output
46	CHG	O	High-side charge drive output
47	VCP	O	Charge pump output, connected to VTOP through a 2.2μF capacitor near the

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Pin	Name	Type	Description
			pin
48	VTOP	S	Power input, connected to positive terminal of pack, connected to VSS through a 1μF capacitor near the pin

Note: I represents input pin, O represents output pin, I/O represents input/output pin, and S represents power pin.

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Absolute Maximum Ratings

Symbol	Parameter	Conditions	Minimum value	Maximum value	Unit
V_{TOP}	VTOP power pin input voltage	VTOP-VSS	-0.3	132	V
V_{CELL}	Cell detection pin input voltage	C0-VSS	-0.3	6	V
		C(n)-VSS, n=1…24	-0.3	132	V
		C(n)-C(n-1), n=1…24	-0.3	132	V
V_{CP}	Output voltage of charge pump pin	VCP-VSS	-0.3	132	V
		VCP-VTOP	-0.3	15	V
V_{SR}	Current detection pin input voltage	SRP-VSS	-0.3	6	V
		SRN-VSS			
V_{DSG}	High-side discharge drive pin output voltage	DSG-VSS	-0.3	132	V
V_{CHG}	High-side charge drive pin output voltage	CHG-VSS	-0.3	132	V
V_{PDSG}	High-side pre-discharge drive pin output voltage	PDSG-VSS	-0.3	132	V
V_{PCHG}	High-side pre-charge drive pin output voltage	PCHG-VSS	-0.3	132	V
V_{PACK}	Charger detection pin voltage	PACK-VSS	-0.3	132	V
V_{LOAD}	Load detection pin voltage	LOAD-VSS	-0.3	132	V
V_{BASE}	NPN Output voltage of base drive pin	VBASE-VSS	-0.3	15	V
		VBASE-VTOP	-132	0.3	V
V_{REG}	5 V power pin input voltage	VREG-VSS	-0.3	6	V
V_{3P3}	3.3V power pin output voltage	V3P3-VSS	-0.3	6	V
		V3P3-VREG	-6	0.3	V
V_{1P8}	1.8V power pin output voltage	V1P8-VSS	-0.3	2	V
		V1P8-VREG	-6	0.3	V
V_{DIG}	Digital port voltage	SCL-VSS	-0.3	6	V
		SDA-VSS			
		GP(n)-VSS, n=1…6			
I_{IO}	Current input / output of the pins	C(n), n=0…24	0	25	mA
		VREG, V3P3	0	50	mA
		other	0	1	mA
T_J	Rated junction temperature range		-40	85	°C
T_S	Storage temperature range		-65	150	°C
V_{ESD}	ESD Rating	HBM mode		±2000	V
		CDM mode		±250	V

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Recommended Operating Conditions

Symbol	Parameter	Conditions	Minimum value	Typical value	Maximum value	Unit
V_{TOP}	VTOP power pin input voltage	VTOP-VSS	8		120	V
		VTOP-C24	-1		120	V
V_{CELL}	Cell detection pin input voltage	C0-VSS	0		0.3	V
		C(n)-C(n-1), n=1…24	0		5.0	
V_{CP}	Output voltage of charge pump pin	VCP-VTOP	0		12	V
		VCP-VSS	0		120	V
V_{SR}	Current detection pin input voltage High-side discharge drive pin output voltage	SRP-VSS	-150		150	mV
		SRN-VSS				
		SRP-SRN				
V_{DSG}	High-side charge drive pin output voltage	DSG-VSS	0		VCP	V
V_{CHG}	High-side pre-discharge drive pin output voltage	CHG-VSS	VTOP		VCP	V
V_{PDSG}	High-side pre-charge drive pin output voltage	PDSG-VSS	0		120	V
V_{PCHG}	Current detection pin input voltage	PCHG-VSS	0		120	V
V_{PACK}	Charger detection pin voltage	PACK-VSS	0		120	V
V_{LOAD}	Load detection pin voltage	LOAD-VSS	0		120	V
V_{BASE}	NPN Output voltage of base drive pin	VBASE-VSS	0		6.5	V
V_{REG}	5V power pin input voltage	VREG-VSS	0		5.5	V
V_{3P3}	3.3V power pin input voltage	V3P3-VSS	0		3.3	V
V_{1P8}	1.8V power pin input voltage	V1P8-VSS	0		1.8	V
V_{DIG}	Digital port voltage	SCL-VSS	0		5	V
		SDA-VSS				
		GP(n)-VSS, n=1…6				
V_{TS}	Input voltage of thermistor pin	GP(n)-VSS, n=1…6	0		1.8	V
V_{AIN}	Analog pin input voltage	GP(n)-VSS, n=1…6	0		1.8	V
I_{CB}	Passive balancing driving current of cell	C(n), n=0…24	0		25	mA
R_C	VADC input filtering			220		Ω

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Symbol	Parameter	Conditions	Minimum value	Typical value	Maximum value	Unit
	resistor					
C_C	VADC input filtering capacitor			0.1		μF
T_{OPR}	Functional temperature		-40		85	°C

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Electrical Characteristics

Symbol	Parameter	Conditions	Minimum value	Typical value	Maximum value	Unit
		VADC Specification				
	input range	C(n)-C(n-1), n=1…24	-0.3		5.0	V
		GP(n)-VSS, n=1…6	0		1.98	V
		V1P8-VSS	0		1.98	V
		C24-VSS, PACK-VSS, LOAD-VSS	0		50	V
	measurement resolution	C(n)-C(n-1), n=1…24		100		μV/bit
		GP(n)-VSS, n=1…6				
		V1P8-VSS				
		C24-VSS, PACK-VSS, LOAD-VSS		12.8		mV/bit
	ADC offset voltage	C(n)-C(n-1), n=1…24		1		mV
		GP(n)-VSS, n=1…6				
		V1P8-VSS				
		C24-VSS, PACK-VSS, LOAD-VSS				
	ADC gain error	C(n)-C(n-1), n=1…24		0.1		%
		GP(n)-VSS, n=1…6				
		V1P8-VSS				
		C24-VSS, PACK-VSS, LOAD-VSS				
	Measurement error at 25°C	C(n)-C(n-1), n=1…24		±5		mV
		GP(n)-VSS, n=1…6				
		V1P8-VSS				
		C24-VSS, PACK-VSS, LOAD-VSS		±500		mV
	Input leakage current without measurement	C(n), n=0…24		±10		nA
		GP(n), n=1…6				
		V1P8				
		C24, PACK, LOAD		10		nA
	Input leakage current with measurement	C(n), n=0…24		±1		μA
		GP(n), n=1…6				
		V1P8				
		C24, PACK, LOAD			31	μA
	measurement time	Single channel	0.791	1.54	6.02	ms
		All channel	34.5	61.4	223	ms
CADC Specification						
	input range		-150		150	mV
	measurement resolution	CC1		5		μV/bit

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Symbol	Parameter	Conditions	Minimum value	Typical value	Maximum value	Unit
		CC2		0.3125		μV/bit
	ADC offset voltage			50		μV
	ADC gain error			0.1		%
	Measurement error			±200		μV
	Input leakage current without measurement	SRP, SRN		±10		nA
	Input leakage current with measurement	SRP, SRN		±1		μA
	measurement time	CC1	0.5		4.0	ms
		CC2		256		ms
General DC specification						
I_{BAT}	VTOP power supply current	Normal mode		15		μA
		Sleep mode		8		μA
		Shutdown mode		1		μA
I_{REG}	VREG power supply current	Normal mode		255		μA
		Sleep mode		52		μA
		Shutdown mode		0.1		μA
LDO Specification						
V_{3P3}	V3P3 output voltage			3.3		V
I_{3p3}	V3P3 output current		0		50	mA
Cell passive balancing drive specification						
R_{CB_ON}	Driven conduction resistance of cell passive balancing	C(n)~C(n-1)，n=4…24	173	290	441	Ω
		C(n)~C(n-1)，n=1…3	123	170	204	Ω
I_{CB_ON}	Driven conduction current of cell passive balancing	C(n)~C(n-1)，n=1…24	0		25	mA
Charge pump Specification						
V_{CP}	Charge pump output voltage	VCP-VTOP	6		12	V
T_{CP_START}	Charge pump start time	C_{CP}=2.2μF, V_{CP}=10V		91		ms
Charge/Discharge drive Specification						
T_{CHG_ON}	Drive rise time when turning on the charge	C_L=47.2nF, R_{GATE}=51Ω, V_{CP}=10V, V_{GS} from 0V to 4V	44			μs
T_{DSG_ON}	Drive rise time when turning on the discharge	C_L=47.2nF, R_{GATE}=51Ω, V_{CP}=10V, V_{GS} from 0V to 4V	26			μs
T_{CHG_OFF}	Drive descent time when shutting down the charge	C_L=47.2nF, R_{GATE}=51Ω, V_{CP}=10V	199			μs

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Symbol	Parameter	Conditions	Minimum value	Typical value	Maximum value	Unit
T_{DSG_OFF}	Drive descent time when shutting down the discharge	C_L=47.2nF, R_{GATE}=51Ω, V_{CP}=10V	6.5			μs
$T_{PCHG_PDSG_ON}$	Drive descent time when turning on the pre-charge / pre-discharge	C_L=860pF, R_{PU}=270kΩ	2.0			ms
$T_{PCHG_PDSG_OFF}$	Drive rise time when shutting down the pre-charge / pre-discharge	C_L=860pF, R_{PU}=270kΩ	2.0			ms
$I_{PCHG_PDSG_ON}$	Pin pulldown current when turning on the pre-charge / pre-discharge		23.2	36.0	54.4	μA
Hardware protection Specification						
V_{OV_TH}	Cell overvoltage protection threshold		501		4595	mV
$V_{OV_TH_STEP}$	Cell overvoltage protection threshold step-by-step			1		mV
T_{OV_DLY}	Cell overvoltage protection delay		200		8000	ms
V_{UV_TH}	Cell undervoltage protection threshold		1		4095	mV
V_{UV_STEP}	Cell undervoltage protection threshold step-by-step			1		mV
T_{UV_DLY}	Cell undervoltage protection delay		200		8000	ms
V_{OC1_TH}	1st level charge/discharge overcurrent protection threshold		0.25		63.75	mV
$V_{OC1_TH_STEP}$	1st level charge/discharge overcurrent protection threshold step-by-step			0.25		mV
T_{OC1_DLY}	1st level charge/discharge overcurrent protection delay		8		2048	ms
$T_{OC1_DLY_STEP}$	1st level charge/discharge overcurrent protection delay step-by-step			8		ms
V_{OC2_TH}	2nd level charge/discharge overcurrent protection threshold		4		256	mV

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Symbol	Parameter	Conditions	Minimum value	Typical value	Maximum value	Unit
$V_{OC2_TH_STEP}$	2nd level charge/discharge overcurrent protection threshold step-by-step			4		mV
T_{OC2_DLY}	2nd level charge/discharge overcurrent protection delay		4		1024	ms
$T_{OC2_DLY_STEP}$	2nd level charge/discharge overcurrent protection delay step-by-step			4		ms
V_{SCD_TH}	Discharge short circuit protection threshold		10		630	mV
$V_{SCD_TH_STEP}$	Discharge short circuit protection threshold step-by-step			10		mV
T_{SCD_DLY}	Discharge short circuit protection delay	Note 1	0		1992	μs
$T_{SCD_DLY_STEP}$	Discharge short circuit protection delay step-by-step			7.81		μs
I²C Interface DC Specification						
V_{IH_I2C}	I²C pin digital input voltage is high level	SCL, SDA	1.25			V
V_{IL_I2C}	I²C pin digital input voltage is low level	SCL, SDA			0.9	V
I_{LEAK_DIG}	Digital input leakage current	SCL, SDA			±1	μA
V_{OL_SDA}	The digital output voltage of SDA pin is low level	SDO Pulldown current 1mA			0.3	V
I²C Timing Specification						
f_{SCL}	SCL clock frequency	The duty cycle is 50%			100	kHz
$t_{HD;STA}$	The holding time of the (repeated) starting condition, after which the first clock pulse is generated		4.0			μs
t_{LOW}	SCL clock low-level cycle		4.7			μs
t_{HIGH}	SCL clock high-level cycle		4.0			μs
$t_{HD;DAT}$	Data retention time		0			ns
$t_{SU;DAT}$	data setup time		250			ns
t_r	Rise time of SCL and SDA	10% to 90%			1000	ns
t_f	Descending time of SCL	90% to 10%			300	ns

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Symbol	Parameter	Conditions	Minimum value	Typical value	Maximum value	Unit
	and SDA					
$t_{SU;STO}$	The establishment time of the stop condition		4.0			μs
t_{BUF}	Bus idle time between stop and start conditions		4.7			μs
R_{PULLUP}	Bus pull-up resistor		1.5			kΩ
t_{TIME_OUT}	Bus timeout time			64		ms

Note 1: This time only refers to the internal logic delay of the chip, and does not include the input RC filter delay and output drive FET load delay.

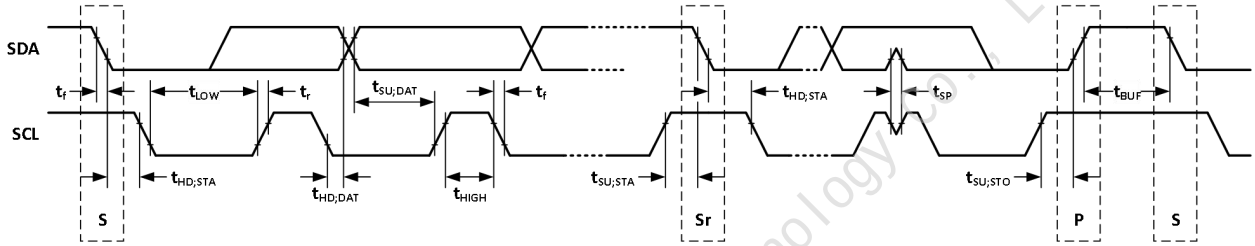


FIG 1. I²C Communication bus timing

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System Block Diagram

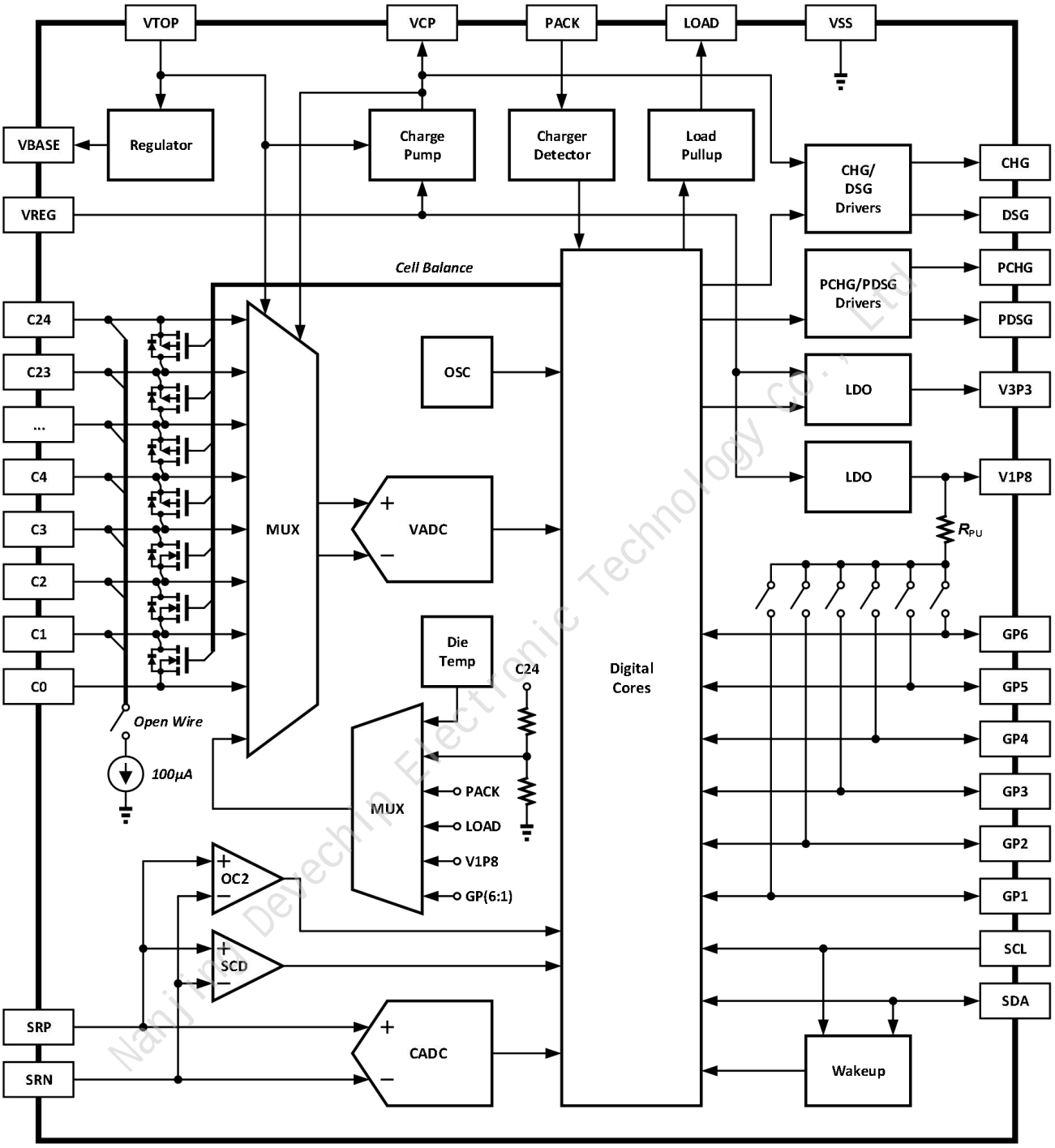


FIG 2. System Block Diagram

Power Modes

DVC1124 has three power modes: shutdown mode, sleep mode, and normal mode.

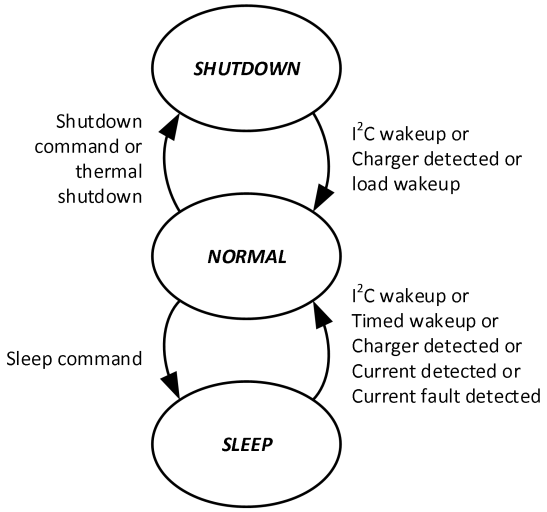


FIG 3 Functional state diagram

Table 1. function list

Mode name	Normal	Sleep	Shut down
VADC	√		
CADC	√		
Charge Pump	√	√	
FET drive	√	√	
balancing	√		
COV/CUV	√		
OCD1/OCC1	√		
OCD2/OCC2	√	√	
SCD	√	√	
VBASE	√	√	
V3P3	√	√	
I²C comm	√		
Current wake		√	
Timer wake		√	
I²C wake		√	√
LOAD wake			√
Charge detection	√	√	√
supply current	~270μA	~60μA	~1μA

Shutdown Mode

In shutdown mode, DVC1124 closes all modules except for I2C wake-up and charger detection modules, and resets the registers to their initial state. There are two ways for chips to enter shutdown mode:

1) Send shutdown command through I2C communication;
2) The chip core shuts down on over temperature.

The chip will exit shutdown mode and enter normal mode in the following two states:

1) Wake up signal detected at I2C communication pin;
2) Apply a voltage of more than 2V higher than the VTOP pin to the PACK pin;
3) Apply a voltage of more than 2V to the LOAD pin.

Sleep Mode

When DVC1124 is in sleep mode, VADC, CADC, COV/CUV protection, and I²C communication functions are turned off. Register data, V3P3, Charge Pump, FET drive, OCD2/OCC2 protection, and SCD protection will remain in the state before entering sleep mode, while enabling timed wake-up and current wake-up detection functions. There is only one way for the chip to enter sleep mode:

1) Send sleep instructions through I²C communication.

The chip will exit sleep mode and enter normal mode in the following 6 states:

1) Wake up signal detected at I²C communication pin;
2) Timed wake-up countdown ends;
3) Detected charging/discharging current;
4) Detected the 2nd level charging/discharging overcurrent (while turning off charging/ discharging drive);

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5) Detected a discharge short circuit (while turning off the discharge drive);
6) Apply a voltage of more than 2V higher than the VTOP pin to the PACK pin.

After the chip returns to normal mode, an interrupt signal can be sent to the MCU through the GP pin. MCU can also query the reason why the chip has exited sleep mode by reading registers.

Normal Mode

In normal mode, DVC1124 supports completed battery pack measurement, protection, and management functions, and MCU can read and configure chip status through I2C communication.

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Measurement system

Current Measurement

DVC1124 integrates 1 Σ-Δ CADC, 2 Coulometer (CC1 and CC2). In normal mode, CADC can continuously measure the voltage difference between SRP and SRN.

CADC Measurement Period

The measurement timing of CC1 and CC2 is shown in FIG 4. The measurement time t_{CM2} for CC2 is fixed at 256ms, and the measurement time t_{CM1} for CC1 is shown in Table 2 under different configurations.

Table 2. CADC CC1 measurement time

C1OW	t_{CM1}	Unit
0x00	0.5	ms
0x01	1.0	ms
0x02	2.0	ms
0x03	4.0	ms

After each CC1 measurement is completed, the chip will perform the following operations:

1) Register CC1F is set to 1.

After each CC2 measurement is completed, the chip will perform the following operations:

1) Register CC2F is set to 1.
2) The Interrupt controller sends a 1ms low-level pulse only once.

CADC Input Range and Resolution

CADC input range is ±150mV. CC1 data format is a 16-bit signed integer, LSB is 5μV；CC2 data format is a 20-bit signed integer， LSB is 0.3125μV.

Voltage Measurement

DVC1124 integrates 1 Σ-Δ VADC, and supports the voltage measurement of C24, PACK, LOAD, Core temperature, V1P8, 6 GPs and 24 Cells. VADC can continuously measure the above voltage in normal mode.

VADC Measurement Period

The VADC measurement cycle consists of 42 time slices, including 36 measurement time slices and 6 delay time slices.

When GP(n) is configured as a thermistor input, VADC will insert a time slice(t_{VD}) delay of 1ms before the corresponding measurement time slice to meet the requirement of external RC establishment time.

Among all 42 time slices, the six measurement time slices(t_{VM1}, t_{VM2} and t_{VM13} to t_{VM16}) cannot be masked and will appear in each VADC measurement cycle. For the 20 measurement time slices(t_{VM17} to t_{VM36}), when the register CMM is set to 0, VADC will skip the measurement time slices corresponding to the masked channels during measurement. When the register CMM is set to 1, VADC will not skip the measurement time slices corresponding to the masked channels during measurement. For the remaining 16 time slices, VADC will skip the time slices corresponding to the masked channels during measurement.

The unit measurement time t_{VM} and the longest measurement cycle t_{VADC} under different configurations of VADC are shown in Table 3:

Table 3. VADC measurement time

VAO	t_{VM}	t_{VADC}	Unit
0x00	0.791	34.5	ms
0x01	1.54	61.4	ms
0x02	3.03	115	ms
0x03	6.02	223	ms

After each VADC measurement is completed, the chip will perform the following operations:

1) Set register VADF to 1;
2) The interrupt controller sends a 1ms low-level pulse once.

VADC Polling Mode

VADC supports both continuous measurement mode and synchronous measurement mode. In continuous measurement mode (register VASM set to 0), VADC will immediately start the next

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measurement cycle after the previous measurement cycle ends. At this time, VADC and CADC are in asynchronous measurement state (as shown in FIG 6).

In synchronous measurement mode (register VASM set to 1), VADC will synchronize with CADC to start measurement. After VADC is completed, it will enter low-power mode and wait for CADC CC2 to be completed before starting the next measurement cycle. The synchronization cycle (register VAMP) can be set to 1, 2, 4, or 8 CADC CC2 measurement cycles (as shown in FIG 7 and FIG 8).

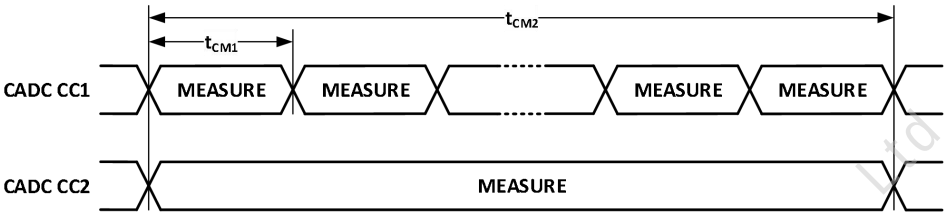


FIG 4. CADC measurement timing

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Cell Voltage Range and Resolution

The cell voltage ranges from -0.3V to +5.0V. When the CVS register is set to 0, the data format is a 16-bit unsigned integer, and the LSB is 100μV. Negative readings are discarded to 0V.

When the CVS register is set to 1, the data format is a 16-bit signed integer, and the LSB is 200μV. Negative readings are valid and retained.

Voltage Range and Resolution of C10, PACK, and LOAD

The voltage ranges for C10, PACK, and LOAD are 0V to 120V, with a data format of a 16-bit unsigned integer and an LSB of 12.8mV.

V1P8 Voltage Range and Resolution

V1P8 voltage ranges from 0V to 1.98V, data format is a 16-bit unsigned integer, LSB is 100μV.

GP Analog Voltage Range and Resolution

DVC1124 can measure up to 6 external analog voltages when GP(n) is configured as an analog voltage input.

GP analog voltage ranges from 0V to 1.98V, data format is a 16-bit unsigned integer, LSB is 100μ V.

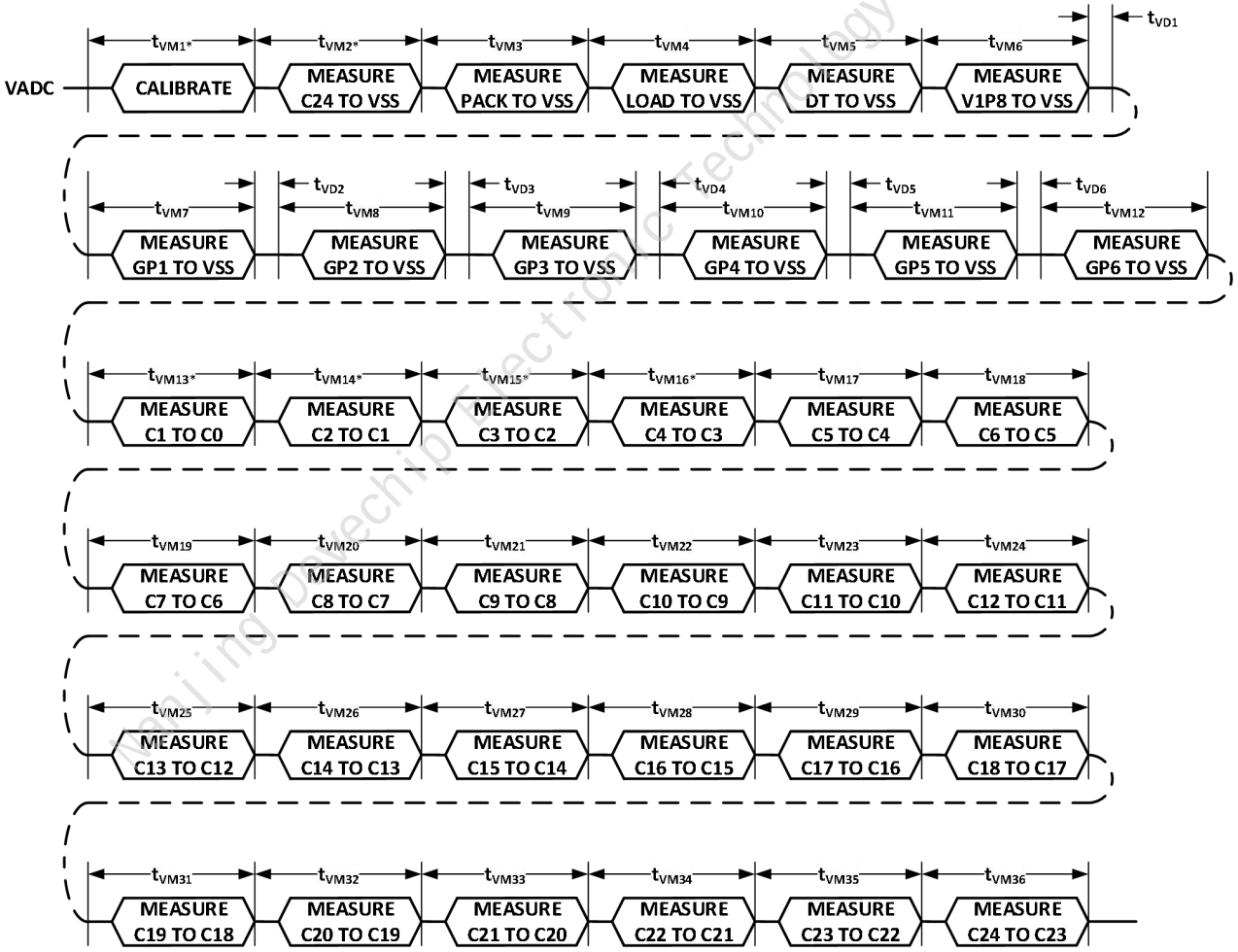


FIG 5. VADC Measurement timing

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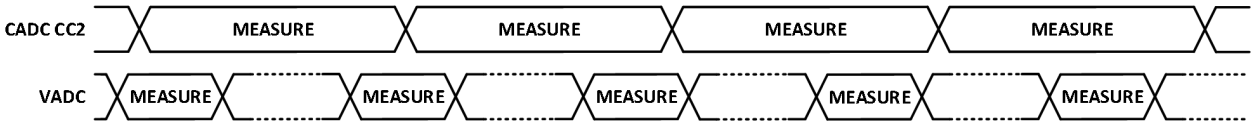


FIG 6. VADC Continuous measurement mode

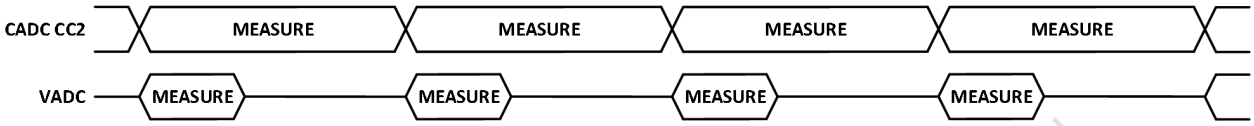


FIG 7. VADC Synchronous measurement mode (1 CC2 cycle)

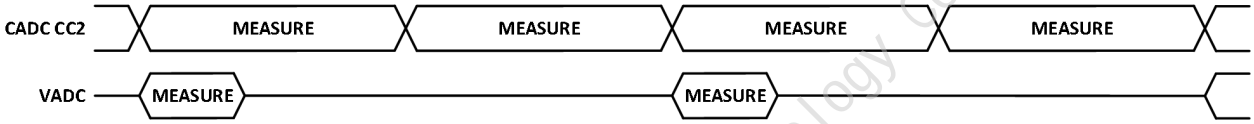


FIG 8. VADC Synchronous measurement mode (2 CC2 cycle)

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Cell Sampling Wiring Method

DVC1124 supports monitoring of 5-series to 24-series battery pack. When the number of cells in series is below 24, the unused cell sampling pins should not be in a suspended state. The sampling pins of C0~C4 pins do not support short circuiting and must be connected to the positive and negative poles of the first to fourth cell strings respectively. There are no restrictions on the sampling pins of other batteries, and suitable short circuits can be adopted according to the sequence of battery pack series connection.

There are two recommended short-circuit methods for DVC1124. The method shown in FIG 9(a) can reduce the influence of the series resistance R_{BUS} between cells, and can measure the cell voltage more accurately during charging and discharging. To simplify the application, the method shown in FIG 9(b) can also be used.

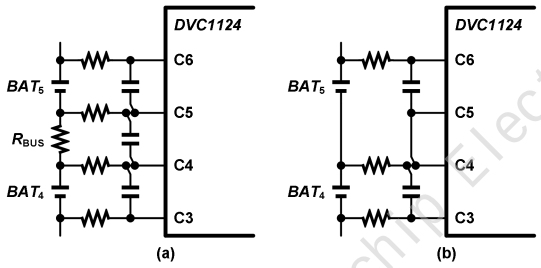


FIG 9. Short circuit of cell sampling

Cell Open Wire Detector

DVC1124 integrates 24 pull-down current sources of 100μA on the sampling pins of C1~C24 cells. When the cell open wire detection function is enabled, the 24 current sources will be pulled down simultaneously. The MCU can determine whether there is a open wire on the cell sampling pin based on the cell voltage measurement value. The time of the timer for the open wire detection function is 1 second. After the countdown ends, the chip will automatically turn off all pull-down current sources.

Temperature Measurement

DVC1124 measures the voltage difference (ΔV_{BE}) between the base and the emitter of an internal transistor through VADC to obtain the core temperature of the chip, which needs to be measured regularly to ensure that the chip does not exceed the rated junction temperature in normal mode.

The calculation formula for chip core temperature (T_{DIE}) is as follows:

$$T_{DIE}=(N_{DT}\times 0.24467)°C - 271.03°C$$

Among them, N_{DT} is the temperature measurement value of the chip core, and the data format is a 16-bit unsigned integer.

Temperature of Thermistor

DVC1124 can measure up to 6 external thermistor temperatures when GP(n) is configured as a thermistor input. Due to the internal pull-up resistance (R_{PU}) having a value of approximately 10kΩ, only thermistors with a resistance value of 10kΩ at room temperature are supported for temperature measurement (FIG 10).

When measuring the temperature of the thermistor, it will first connect the thermistor to the internal pull-up resistor, waiting for 1ms for VADC, and then starting voltage measurement. After the measurement is completed, disconnect the thermistor from the internal pull-up resistor.

When measuring the temperature of a thermistor, a filtering capacitor (C_F) needs to be placed near the chip pins. The selection of filter capacitor capacitance value should consider the influence of RC establishment time, and the calculation formula is:

$$C_F=\frac{1m}{10\times R_{PU}}$$

The recommended capacitance value for filter capacitors (C_F) is generally recommended to be 10nF, which needs to be appropriately reduced when the connection line is long.

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The formula for calculating the resistance value of a thermistor (R_{NTC}) is as follows:

$$R_{NTC}=\frac{N_{VGP}}{N_{V1P8}-N_{VGP}}\times R_{PU}$$

Among them, N_{VGP} is the voltage measurement value of the GP pin, and the data format is a 16-bit unsigned integer; N_{V1P8} is the voltage measurement value of the V1P8 pin, and the data format is a 16-bit unsigned integer.

The internal pull-up resistance value (R_{PU}) will be adjusted during chip final testing, and the calculation formula is as follows:

$$R_{PU}=N_{FRT}\times 25\Omega+6800\Omega$$

Among them, N_{FRT} is an 8-bit unsigned integer.

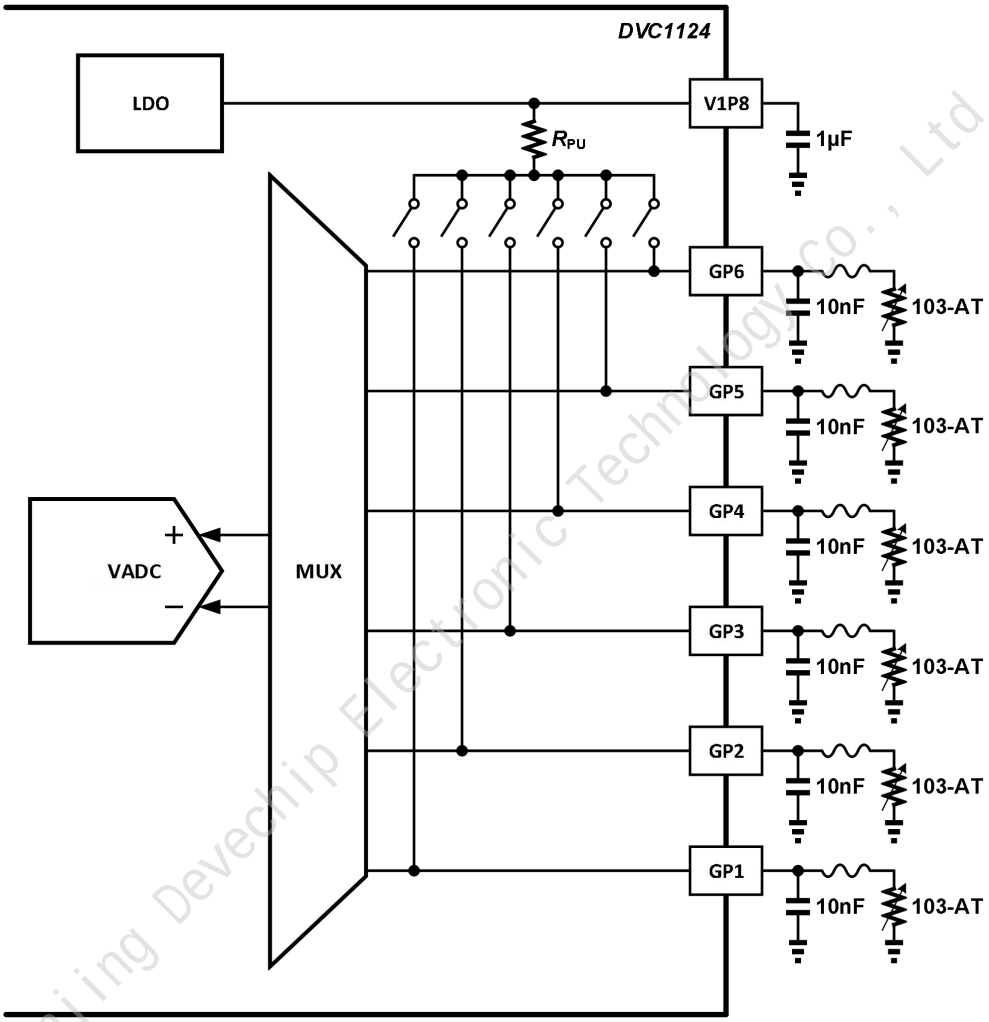


FIG 10. Temperature measurement of thermistors

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Protection System

Cell Voltage Protection

DVC1124 provides cell overvoltage and undervoltage protection based on VADC measurement values.

Cell Overvoltage Protection (COV)

To activate cell overvoltage protection, the following conditions must occur simultaneously:

1) The chip is in normal mode;

2) VADC enable control bit VAE is set to 1;

3) The cell overvoltage protection threshold COVT is not 0; the threshold voltage for cell overvoltage protection (V_{COV}) ranges from 501mV to 4595mV, with a step of 1mV; The delay time (T_{COV}) ranges from 200ms to 8s.

The chip integrates 24 ways of cell overvoltage protection timers. When one of the series of cell voltage measurements is greater than V_{COV}, the cell overvoltage protection timer starts. If any voltage measurement value of the cells series during the T_{COV} time is less than V_{COV}, the timer will reset. Otherwise, the countdown will continue. When the countdown of any cell overvoltage protection timer ends, a cell overvoltage protection alarm will be triggered.

The 5th to 24th cells series can be set to a masked state. In the masked state, the overvoltage protection timer of this cell is disabled, and any voltage measurements of the series cells will not trigger a cell overvoltage protection alarm.

After the cell overvoltage protection alarm is triggered, the chip will perform the following actions:

1) Register COV is set to 1;

2) The interrupt controller sends an one-shot 1ms low-level pulse;

After the cell overvoltage protection alarm is triggered, the register COV will remain locked until any of the following conditions are triggered to release this alarm:

1) Register COV is reset 0;

2) VADC enable control bit VAE is set to 0;

3) The chip is in sleep mode.

Cell Undervoltage Protection (CUV)

To activate cell undervoltage protection, the following conditions must occur simultaneously:

1) The chip is in normal mode;

2) VADC enable control bit VAE is set to 1;

3) The cell undervoltage protection threshold CUVT is not 0; the threshold voltage for cell undervoltage protection (V_{CUV}) ranges from 1mV to 4095mV in steps of 1mV. The delay time (T_{CUV}) ranges from 200ms to 8s.

The chip integrates 24 cell undervoltage protection timers. When one of the cell voltage measurements is less than V_{CUV}, the undervoltage protection timer for that cell will starts. If any voltage measurement value of the series of batteries within the set T_{CUV} is greater than V_{CUV}, the timer will reset. Otherwise, the countdown will continue. When the countdown of any cell undervoltage protection timer ends, a cell undervoltage protection alarm will be triggered.

The 5th to 24th series of batteries can be set to a masked state. In the masked state, the undervoltage protection timer for this cell is disabled, and any measurement of the voltage of this series of cells will not trigger a cell undervoltage protection alarm.

After the cell undervoltage protection alarm is triggered, the chip will perform the following actions:

1) Register CUV is set to 1;

2) The interrupt controller sends an one-shot 1ms low-level pulse;

After the cell undervoltage protection alarm is triggered, the register CUV will remain locked

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until any of the following conditions are triggered to release the alarm:

1) Register CUV is set to 0;

2) VADC enable control bit VAE is set to 0;

3) The chip is in sleep mode.

Charge/discharge Current Protection

DVC1124 provides 1st level charge/discharge overcurrent protection based on the values of CADC measurement, 2nd level charge/discharge overcurrent protection based on hardware overcurrent comparator (OC2), and discharge short-circuit protection based on hardware short-circuit comparator (SCD).

1st discharge overcurrent protection (OCD1)

To activate the first level discharge overcurrent protection, the following conditions must occur simultaneously:

1) The chip is in normal mode;

2) CADC enable control bit CAE is set to 1;

3) The first level discharge overcurrent protection threshold OCD1T is not 0; the threshold voltage range for the first level discharge overcurrent protection (V_{OCD1_TH}) is 0.25mV to 63.75mV, with a step of 0.25mV; The delay time (T_{OCD1_DLY}) ranges from 8ms to 2048ms, with a step of 8ms.

When the voltage measurement difference of SRP-SRN is greater than V_{OCD1_TH}, the first level discharge overcurrent protection timer starts. At the set T_{OCD1_DLY}, if any SRP-SRN voltage measurement difference is less than V_{OCD1_TH}, the timer will reset, otherwise the countdown will continue. At the end of the countdown, the first level discharge overcurrent protection alarm is triggered. The chip performs the following operations:

1) Register OCD1 is set to 1;

2) The interrupt controller sends an one-shot 1ms low-level pulse.

After the first level discharge overcurrent protection alarm is triggered, register OCD1 will remain locked until any of the following conditions are triggered to release this alarm:

1) Register OCD1 is reset to 0;

2) CADC enable control bit CAE is set to 0;

3) The chip is in sleep mode.

1st Charge Overcurrent Protection (OCC1)

To activate the first level charging overcurrent protection, the following conditions must occur simultaneously:

1) The chip is in normal mode;

2) CADC enable control bit CAE is set to 1;

3) The first level charging overcurrent protection threshold OCC1T is not 0; the threshold voltage for overcurrent protection of the first level of charging (V_{OCC1_TH}) ranges from 0.25mV to 63.75mV, with a step of 0.25mV; The delay time (T_{OCC1_DLY}) ranges from 8ms to 2048ms, with a step of 8ms.

When the difference of SRN-SRP voltage measurement is greater than V_{OCC1_TH}, the first level charging overcurrent protection timer starts. At the set V_{OCC1_TH}, if any SRN-SRP voltage measurement difference is less than V_{OCC1_TH}, turn off the timer, otherwise the countdown will continue. At the end of the countdown, the first level charging overcurrent protection alarm is triggered. The chip performs the following operations:

1) Register OCC1 is set to 1;

2) The interrupt controller sends a 1ms low-level pulse once;

After the 1st level charging overcurrent protection alarm is triggered, register OCC1 will remain locked until any of the following conditions are triggered to release this alarm:

1) Register OCC1 is set to 0;

2) CADC enable control bit CAE is reset to 0;

3) The chip is in sleep mode.

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2nd Discharge Overcurrent Protection (OCD2)

To activate the 2nd level discharge overcurrent protection, the following conditions must occur simultaneously:

1) The chip is in normal mode or sleep mode;

2) OC2 enable control bit OCD2E is set to 1;

The threshold voltage for the 2nd level discharge overcurrent protection (V_{OCD2_TH}) ranges from 4mV to 256mV, with a step of 4mV; The delay time (T_{OCD2_DLY}) ranges from 4ms to 1024ms, with a step of 4ms.

When the difference in voltage measurement between SRP and SRN is greater than V_{OCD2_TH}, the second level discharge overcurrent protection timer starts. In the set T_{OCD2_DLY}, if any SRP-SRN voltage measurement difference is less than V_{OCD2_TH}, turn off the timer, otherwise the countdown will continue. At the end of the countdown, the 2nd level discharge overcurrent protection alarm is triggered. The chip performs the following operations:

1) Set register OCD2 to 1;

2) The interrupt controller to send an one-shot 1ms low-level pulse;

After the second level discharge overcurrent protection alarm is triggered, register OCD2 will remain locked until any of the following conditions are triggered to release this alarm:

1) Register OCD2 is reset to 0;

2) Reset OC2 enable control bit OCD2E to 0.

2nd Charge Overcurrent Protection (OCC2)

To activate the second level charging overcurrent protection, the following conditions must occur simultaneously:

1) The chip is in normal mode or sleep mode;

2) OC2 enable control bit OCC2E is set to 1;

The threshold voltage for the second level charging overcurrent protection (V_{OCC2_TH}) ranges from 4mV to 256mV, with a step of 4mV; The delay time (T_{OCC2_DLY}) ranges from 4ms to 1024ms, with a step of 4ms.

When the voltage difference measurement between SRP and SRN is greater than V_{OCC2_TH}, the second level charging overcurrent protection timer starts. If any one of the voltage difference measurements between the SRP and SRN is less than V_{OCC2_TH} within the set T_{OCC2_DLY} time, the timer will turn off, otherwise the countdown continues. At the end of the countdown, the 2nd level charging overcurrent protection alarm will be triggered. The chip performs the following operations:

1) Register OCC2 is set to 1;

2) Interrupt the controller to send one-shot 1ms low-level pulse;

After the second level charging overcurrent protection alarm is triggered, register OCC2 will remain locked until any of the following conditions are triggered to release this alarm:

1) Reset register OCC2 to 0;

2) Reset OC2 enable control bit OCC2E to 0.

Discharge Short Circuit Protection(SCD)

To enable discharge short circuit protection, the following conditions must occur simultaneously:

1) The chip is in normal mode or sleep mode;

2) SCD enable control bit SCDE is set to 1.

The threshold voltage for discharge short circuit protection (V_{SCD_TH}) ranges from 10 mV to 630 mV, with a step of 10mV; The delay time (T_{SCD_DLY}) ranges from 0μs to 1992μs. Step by 7.81μs.

When the voltage difference measurement between SRP and SRN is greater than V_{SCD_TH}, the discharge short circuit protection timer starts.

If any voltage difference measurements between the SRP and SRN is less than V_{SCD_TH} within the set T_{SCD_DLY} time, the timer will turn off, otherwise the countdown continues. At the end of the countdown, a discharge short circuit protection

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alarm will be triggered. The chip performs the following operations:

1) Set register SCD to 1;

2) The Interrupt controller sends one-shot 1ms low-level pulse;

After the discharge short circuit protection alarm is triggered, the register SCD will remain locked until the following conditions are triggered to release the alarm:

1) Reset register SCD to 0.

2) Reset SCD enable control bit SCDE to 0.

I²C Watchdog

To activate the I²C watchdog timer, the following conditions must occur:

1) The chip is in normal mode;

2) I²C watchdog timer control bit IWT is not 0x00;

When the I²C watchdog timer is turned on, if the chip does not receive a valid I²C read or write instruction within the time set in register IWT, it will trigger an I²C watchdog timeout alarm. The chip will perform the following operations:

1) Register IWTS set to 1;

2) Register IWTF is set to 1.

After the I²C watchdog timeout alarm is triggered, register IWTS will remain locked until the following conditions are triggered to release the I²C watchdog timeout alarm:

1) The chip has received valid I²C read or write instruction.

Note that only when the host computer actively resets will the register IWTF value be 0.

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Management System

Cell Balancing Management

DVC1124 integrates 24-way passive cell balancing drivers, which can only be turned on when the chip is in normal mode and the VADC is in low power mode.

Internal MOS Balancing of the Cell

When DVC1124 uses on-chip MOS for balancing (FIG 11), the maximum balancing current of each battery cell in series should not exceed 25mA, and real-time monitoring of the chip core temperature is required to prevent the junction temperature from exceeding the maximum rated value.

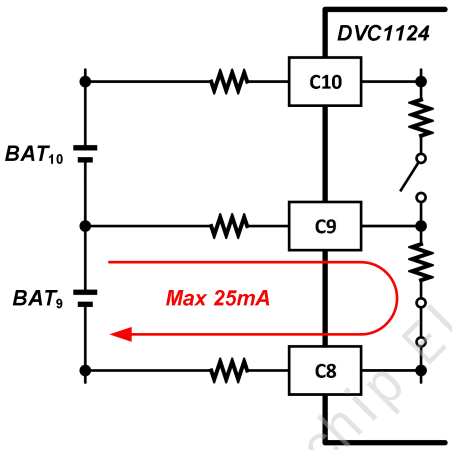


FIG 11. Internal balancing of the cell

External NPN Balancing of the Cell

When DVC1124 uses on-chip MOS to drive off-chip NPN for balancing (FIG 12), the balancing current is not limited by chip heat dissipation and can achieve high-current cell balancing.

Cell Parity Balancing

DVC1124 will perform odd-even series alternating balancing on batteries to avoid the situation of adjacent series cells being balanced simultaneously (FIG 13). The formula for calculating the cell equilibrium time (t_{CB}) is as follows:

$$t_{CB}=N\times t_{CM2}-t_{VADC}$$

Among them, N is the VADC synchronous measurement cycle, which can be 1, 2, 4, or 8; t_{CM2} is the CADC CC2 measurement time, fixed at 256ms; t_{VADC} is the measurement time of VADC, as shown in Table 3.

The chip will sort the battery pack based on the shielding status of the cell sampling pins, starting from the 1st string of cells, and the passive balancing driver of the masked pins will be disabled. Table 4 lists a shielding setting and cell parity grouping applied to 13-cell series battery pack.

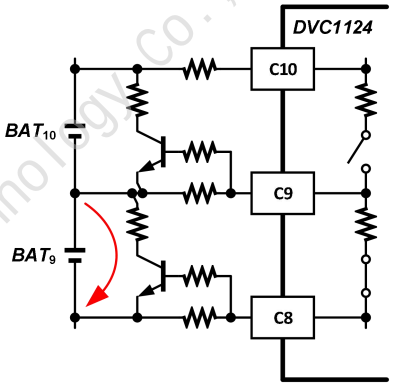


FIG 12. External cell balancing

Table 4. 13 series cell mask settings and parity grouping

Pin	order	mask	odd/even
C1	1	N/A	odd
C2	2	N/A	even
C3	3	N/A	odd
C4	4	N/A	even
C5		yes	N/A
C6	5	no	odd
C7		yes	N/A
C8		yes	N/A
C9	6	no	even
C10	7	no	odd
C11		yes	N/A
C12		yes	N/A
C13		yes	N/A
C14	8	no	even
C15	9	no	odd
C16	10	no	even
C17		yes	N/A
C18		yes	N/A

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C19		yes	N/A
C20		yes	N/A
C21	11	no	odd
C22	12	no	even
C23	13	no	odd
C24		yes	N/A

Cell Balancing Timer

DVC1124 has a built-in passive balancing driver timer, which can be activated when the register control bit of any cell in series is set to 1. Within 60 seconds, if the balancing control bit of any other cell is set to 1, the timer will be reloaded to 60 seconds, otherwise the countdown continues. After the countdown ends, the chip will reset all cell balancing control bits to 0.

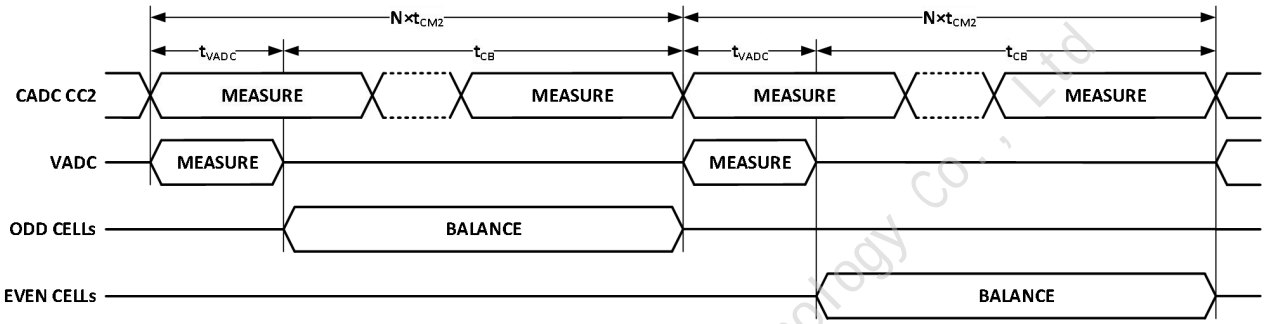


FIG 13. Cell balancing timing

Discharge Management

Charge Pump

DVC1124 integrates a charge pump on-chip and needs an external capacitor located between the VTOP and VCP pins to store charges. When the charge pump is turned on, the capacitor will be charged to an overdrive voltage higher than the VTOP voltage.

The overdrive voltage generated by the charge pump can be used to drive DSG to turn on high-side discharge NFET and drive CHG to turn on high-side charging NFET, and drive MUX to measure cell voltage.

The overdrive voltage can be set between the voltages of 6V and 12V. The higher the overdrive voltage, the more current will be consumed. An appropriate overdrive voltage can be set according to the characteristics of NFET to save current.

High-side Discharge Driver

DVC1124 integrates a high-side discharge driver on-chip, which is used to drive DSG to control high-side discharge NFET. When the high-side discharge NFET is turned off, the high-side discharge driver will pull the DSG down to the GND level. When the high-side discharge NFET is turned on, the high-side discharge driver has two output modes: charge pump output mode and source following output mode.

In the charge pump output mode, the high-side discharge driver will pull the DSG up to the VCP level, and the over driving voltage output by the charge pump will lower the conduction resistance of the discharged NFET.

In the source following output mode, the high-side discharge driver will pull the DSG up to the VTOP level, and the high-side NFET will work in the source following state. The source output mode is suitable for keeping the discharge NFET on while the chip is in sleep mode. At this time, the charge pump can be turned off, reducing the chip's power consumption to a lower level.

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High-side Pre-discharge Driver

DVC1124 integrates a high-side pre-discharge driver, which is used to drive the PDSG to control the high-side pre-discharge PFET. When the pre-discharge PFET is turned off, the PDSG output is in a high resistance state, and the V_{GS} of the pre-discharge PFET is pulled up to 0V by an external 270kΩ resistor. When the pre-discharge PFET is turned on, the PDSG pulls down approximately 36μA current, and the V_{GS} of the pre-discharge PFET is pulled down to approximately 9.7V through an external 270k Ω resistor.

High-side Load Detection

DVC1124 has a built-in high-side load pull-up driver. When the high-side discharge NFET is turned off, turning on the high-side load pull-up driver will output 150μA current and pull the LOAD pin up to the higher level between VTOP and PACK. The MCU can determine the load connection status by reading the load pin voltage through VADC.

The high-side load pull-up driver has a timed shutdown function. Control position 1 of the high-side load pull-up driver and start the timer for 60 seconds. After the countdown ends, the chip will automatically reset the control position of the high-side load pull-up driver to 0.

Low-side Discharge Driver

When the register bit GP6M is set to 0x07, the GP6 pin drives output for low-side discharging.

Low-side Pre-discharge Driver

When the register bit GP2M is set to 0x07, the GP2 pin drives output for low-side pre-discharging.

Discharge Hard Wire Control

When register GP4M is set to 0x03, pin GP4 is set as the discharge hard wire control input. When the GP4 input is low, the discharge drive output is turned off, and when it is high, it does not affect the discharge drive output status.

Charge Management

High-side Charge Driver

DVC1124 integrates a high-side charging driver internally, which is used to drive CHG to control the high-side charging NFET. When the high-side charging NFET is turned off, the high-side charging driver will pull the CHG down to the VTOP level. When the high-side charging NFET is turned on, the high-side charging driver will pull the CHG up to the VCP level.

High-side Pre-charge Driver

DVC1124 integrates a high-side pre-charging driver, which is used to drive PCHG to control the high-side pre charging PFET. When the pre-charged PFET is turned off, the PCHG output is in a high resistance state, and the VGS of the pre-charged PFET is pulled up to 0V by an external 270kΩ resistor. When the pre-charging PFET is turned on, the PCHG pulls down approximately 36μA current, and the V_{GS} of the pre-charged PFET is pulled down to 9.7V approximately through an external 270kΩ resistor.

High-side Charger Detection

DVC1124 has a built-in high-side charger detector. When the PACK level is about 2V higher than the VTOP level, the charger can be determined insert.

Low-side Charge Driver

When the register bit GP5M is set to 0x07, the GP5 pin drives output for low-side charging.

Low-side Pre-charge Driver

When the register bit GP3M is set to 0x07, the GP3 pin drives output for low-side pre-charging.

Charge Hard Wire Control

When register GP1M is set to 0x03, the GP1 pin is the charging-hardwire control input mode. In this case, when the GP1 input is low, the charge drive output will be turned off, and when the

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input is high, the charge drive output state will not be affected.

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Communication System

I²C Serial Interface

The I2C serial interface of DVC1124 operates in slave mode, supporting 100kHz communication rate and CRC8 verification. SCL is a unidirectional clock input pin without clock extension function; SDA is a bidirectional data input/output pin. SCL and SDA are not equipped with pull-up resistors inside the chip and can be compatible with external 5V pull-up levels.

I²C Bus Protocol

The I2C bus generates a clock pulse for every data bit transmitted. The level on the SDA data line must remain stable when the SCL clock is high, and the high and low levels on the SDA data line can only be switched when the SCL clock is low.

When the SCL clock line is at a high level, the SDA data line switches from a high level to a low level, which is defined as the starting condition (S). When the SCL clock line is at high level, the SDA data line switches from low level to high level, which is defined as the stop condition (P). The start and stop conditions are both sent by the host, and the I2C bus is in a busy state after the start condition and in an idle state after the stop condition. If a repeat start condition (Sr) is generated without a stop condition, the bus will remain busy, and the start condition and repeat start condition are functionally the same. In this document, unless otherwise stated Sr, the symbol S will be used as a general term to represent both the starting condition and the repeating starting condition.

Each packet sent to the SDA data line must contain 1 byte and 1 response bit, with each byte first transmitted as the highest bit (MSB). Unlike the standard I2C protocol, even if the I2C clock frequency exceeds the DVC1124 response limit, DVC1124 will not force the host to enter a waiting state by keeping the SCL clock line low.

I²C Slave Address

In typical application scenarios, DVC1124 has a fixed slave address of 0x40 (write) or 0x41 (read) in I2C communication.

For cascading application scenarios, when pins SRN and SRP are short circuited to pins VSS and VREG respectively, pins GP1 and GP2 are fixed in digital input mode. The slave address of DVC1124 in I2C communication is determined by the logical input levels of pins GP1 and GP2 (Table 5).

Table 5. Cascade application I2C slave address

Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1	1	0	0	0	GP2	GP1	R/W

CRC8 Check

The CRC8 verification polynomial is x^8+x^2+x+1, with an initial value of 0x00.

I²C Register Address

The register address of DVC1124 is from 0x00 to 0x90, with a total of 145 bytes. There is no limit on the number of bytes that can be read and written continuously. When the register address reaches the boundary 0x90, it will return to 0x00 to continue reading and writing.

I²C Write Operation

The I2C write operation uses the starting condition sent by the host as the starting signal. The 1st byte transmitted by the host is SA+$\overline{W}$. If SA matches the slave address of DVC1124, DVC1124 will return ACK; Otherwise, NACK will be returned and this transmission ends. The 2nd byte transmitted by the host is RA. If RA is within the address range of 0x00 to 0x8F, DVC1124 will return ACK; Otherwise, NACK will be returned and this transmission ends. The 3rd byte transmitted by the host is DATA0, and DVC1124 will return ACK. The 4th byte transmitted is CRC0. If CRC0 matches the CRC8 checksum of the first three bytes SA+$\overline{W}$, RA, and DATA0, DVC1124 will write DATA0 to the byte at register address RA and return ACK; Otherwise, DATA0

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will be discarded, NACK will be returned and this transmission ends. The 5th byte transmitted by the host is DATA1, and DVC1124 will return ACK. The 6th byte transmitted by the host is CRC1. If the CRC8 checksum of CRC1 matches that of DATA1, DVC1124 will write DATA1 to the byte with register address RA+1 and return ACK; Otherwise, DATA1 will be discarded and NACK will be returned and this transmission ends. And so on until the host sends a stop condition to end this transmission.

I²C Read Operation

The I2C read operation uses the starting condition sent by the host as the starting signal. The 1st byte transmitted by the host is SA+$\overline{W}$. If SA matches the slave address, DVC1124 will return ACK; Otherwise, NACK will be returned and this transmission ends. The 2nd byte transmitted by the host is RA. If RA is within the address range of 0x00 to 0x8F, DVC1124 will return ACK; Otherwise, NACK will be returned and this transmission ends. The 3rd byte transmitted by the host after sending the repeated start condition is SA+R. If SA matches the DVC1124 slave address, DVC1124 will return ACK; Otherwise, NACK will be returned and this transmission ends. The 4th byte transmitted is the value DATA0 of RA returned by DVC1124 and the host needs to respond to ACK. The 5th byte transmitted is the CRC0 returned by DVC1124. If the CRC0 matches the CRC8 checksum of the first four bytes SA+$\overline{W}$, RA, SA+R, and DATA0, it indicates that DATA0 has been read correctly. The host needs to respond to ACK to continue reading. The 6th byte transmitted is the value of the DVC1124's DATA1 returned by the register address RA+1, and the host needs to return ACK. The 7th byte transmitted is the CRC1 returned by DVC1124. If the CRC1 checksum matches the CRC8 checksum of DATA1, it indicates that DATA1 has read correctly. And so on, until the host gives up reading and returns NACK, and finally the host sends a stop condition to end the transfer.

I²C Bus timeout timer

DVC1124 has a built-in bus timeout timer, which starts when it receives the starting condition (excluding duplicate starting conditions) sent by the host. Within 64ms, if a stop condition is received from the host, the timer will be turned off; Otherwise, the countdown will continue. After the countdown ends, the I2C serial interface of DVC1124 will forcibly end the transmission and no longer respond to any signals from the host other than the start condition.

Interrupt Controller

DVC1124 can set GP2, GP3, GP5, or GP6 as interrupt signal output pins. When the chip is in normal mode, by setting the corresponding mask register, one or more trigger events in Table 6 can be selected, and a 1ms low-level pulse can be output through the interrupt signal output pin.

Table 6. Interrupt trigger event

Trigger Event	Mask Register
The chip transitions from sleep mode to normal mode	IWM
VADC completes 1 measurement cycle	IVOM
CADC CC2 completes 1 measurement cycle	ICCM
Trigger cell overvoltage alarm	ICOM
Trigger cell undervoltage alarm	ICUM
Trigger 1st level overcurrent alarm	IOC1M
Trigger 2nd level overcurrent alarm	IOC2M
Trigger discharge short circuit alarm	ISCDM

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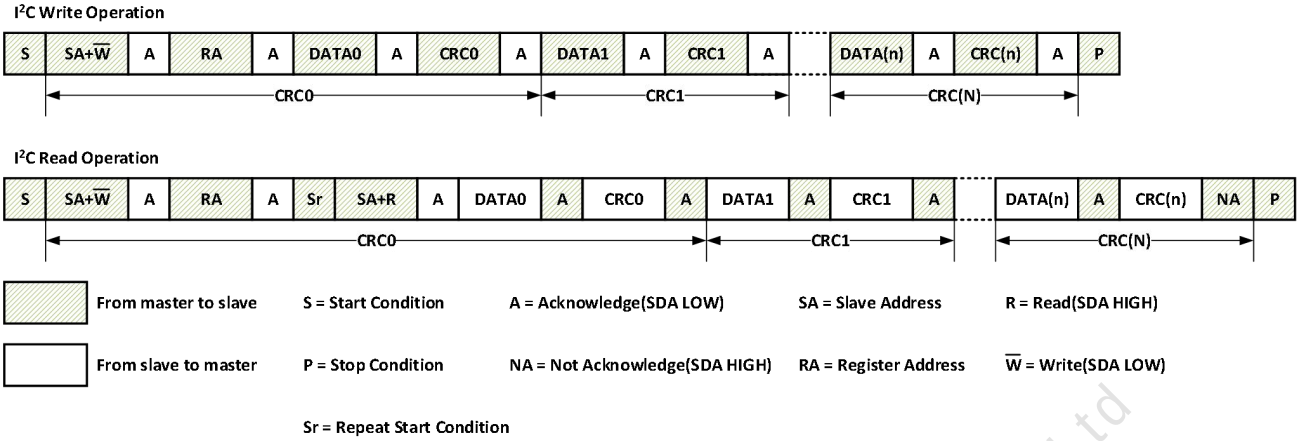


FIG 14. I²C Transport Protocol

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Application and Implementation

Conformal Coating

To meet the requirements of IPC-2221B or IEC/UL 60950-1 for the electrical clearance and creepage distance between adjacent pins, DVC1124 may need to be covered with conformal coatings during the assembling process of printed circuit boards.

Unused Pins

Table 7. Unused pin handling method

High / Low-side drive	High-side		Low-side
Same/Separate Application	Same	Separate	Same/ separate
C5~C24	When not in use, refer to the "Cell sampling wiring method" to short-circuit to adjacent pin		
SRP, SRN	Short circuited to pin VSS when not in use		
GP1~GP6	Be suspended when not in use		
V3P3	When not in use, it can be suspended or passed through 1μF Connect the capacitor to pin VSS and set registers V3P3EW and V3P3ES to 0		
DSG, CHG	Be suspended when not in use		Should be suspended
PDSG, PCHG	Short circuited to pin VSS when not in use		Short circuited to pin VSS
LOAD	When pin DSG is not in use, pin LOAD can be connected to the PACK+ of battery pack through a 10kΩ resistor or short circuited to pin VSS		Short circuited to pin VSS
PACK	When pin CHG is not in use, pin PACK can be connected to the PACK+ of battery pack through a 10kΩ resistor or short circuited to pin VSS	When pin CHG is not in use, pin PACK can be connected to the CPACK+ of the pack through a 10kΩ resistor or short circuited to pin VSS	Short circuited to pin VSS
VCP	When both pins DSG and CHG are not in use, a 100nF capacitor should be connected to the VTOP near the chip's pins, and the register CPVS should be set to 0x01		It should be connected to VTOP through a 100nF capacitor near the chip pins, and register CPVS should be set to 0x01

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Recommended Layout

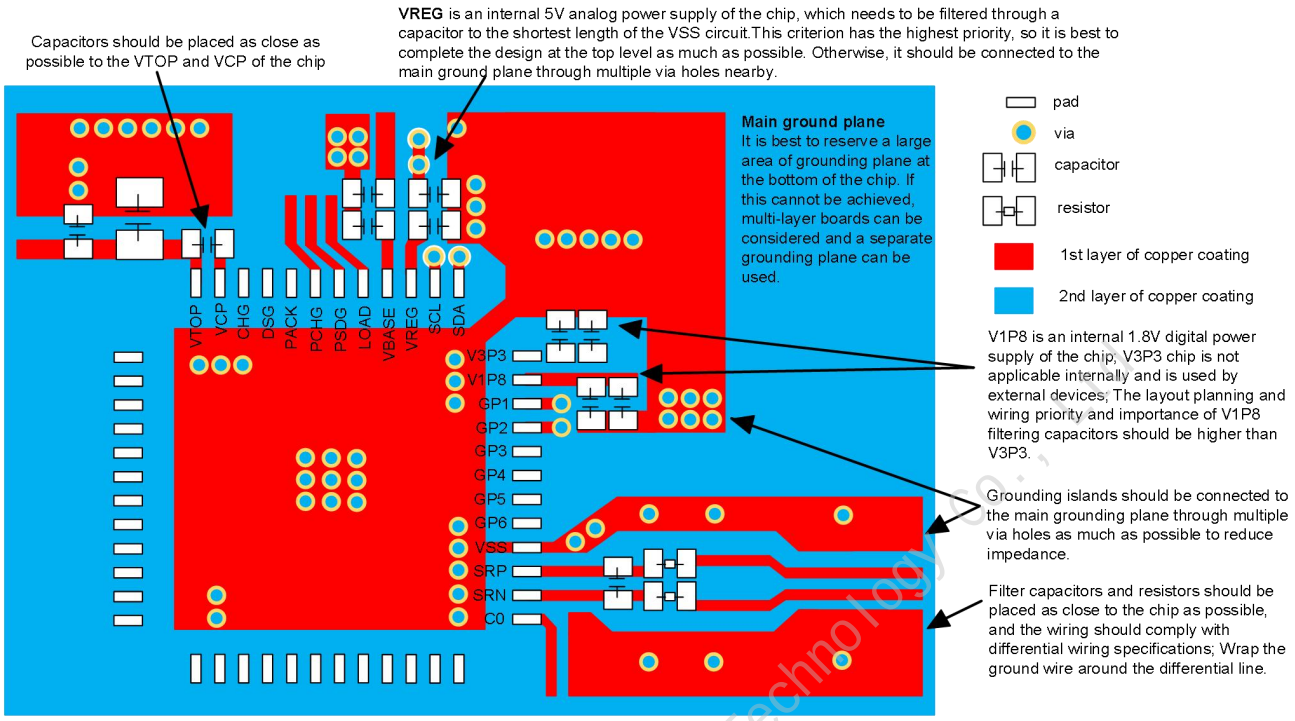


FIG 15. PCB Recommended layout

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Application Block Diagram

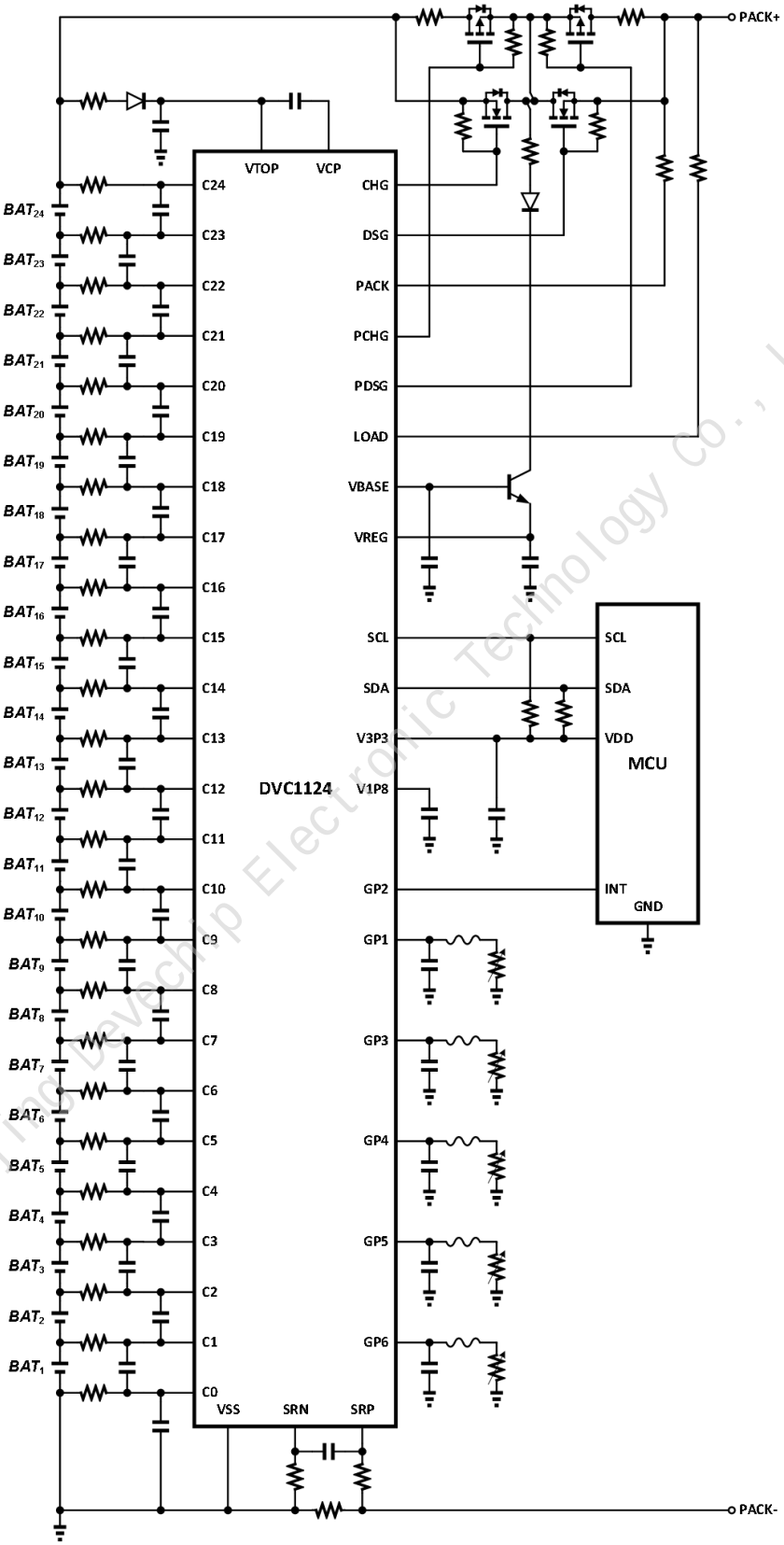
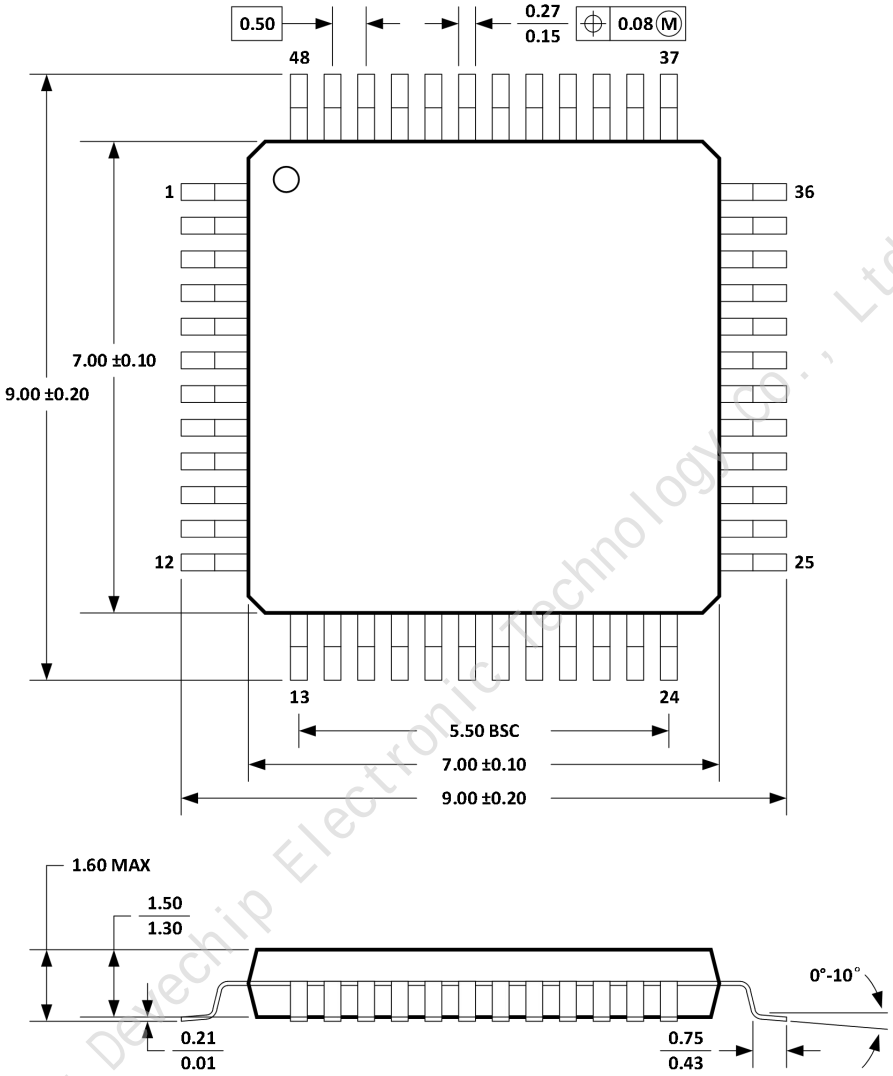


FIG 16. Application Block Diagram

Package Specification

LQFP48(7mm×7mm)



Note: All linear dimensions are in millimeters

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Ordering Information

Device marking	Packaging description	MSL Grade	Packaging method	Minimum order unit
DVC1124-12	48-Lead Plastic LQFP	3	Tray(250pcs)	2500pcs

Revised History

edition	date	description	page
1.0	2023/4/2	Modify "DSG and CHG Absolute Maximum Rating"	8
		Modify "DSG and CHG recommended operating voltage"	9
		Modify "conditions for cell overvoltage protection" and "conditions for removing cell overvoltage protection alarm"	25
		Modify the conditions for opening the cell undervoltage protection and the conditions for removing the cell undervoltage protection alarm	25
		Modify "conditions for opening level 1 discharge overcurrent protection" and "conditions for removing level 1 discharge overcurrent alarm"	26
		Modify "conditions for opening level 1 charging overcurrent protection" and "conditions for removing level 1 charging overcurrent alarm"	26
		Modify "conditions for opening level 2 discharge overcurrent protection" and "conditions for removing level 2 discharge overcurrent alarm"	27
		Modify "conditions for opening stage 2 charge overcurrent protection" and "conditions for lifting stage 2 charge overcurrent alarm"	27
		Modify the “High-side pre-discharge driver” description	31
		Modify the “High-side pre-charge driver" description	31
		revise "I^2C slave address" description	33
		revise "I^2C register address" description	33
		Add "Application and Implementation"	36

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